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**ADVANCED MACHINE LEARNING:
CONCRETE COMPRESSIVE STRENGTH PREDICTION**

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1 Introduction

The objective of this project is to model and predict the compressive strength of concrete, which is a highly complex and non-linear function of its age and ingredients. Concrete is the most critical material in civil engineering. Accurately estimating its strength using machine learning models provides immense value, as it can save significant time and resources compared to traditional physical tests that often require waiting up to 28 days.

1.1 Dataset presentation

The dataset used for this study, the *Concrete Compressive Strength* dataset, consists of 1030 observations. This task is framed as a non-linear regression problem. A defining characteristic of this dataset is that all independent variables (Table 1), as well as the target variable (Table 2), are strictly continuous numerical values. Consequently, no categorical encoding is required. However, the variables represent different physical measurements (e.g., kilograms per cubic meter for ingredients, and days for age), making feature scaling an essential preprocessing step.

Table 1: Description and domains of the dataset’s continuous variables

Feature	Description	Domain
Cement	Amount of cement (kg in a m ³ mixture)	[102.0, 540.0]
BlastFurnaceSlag	Amount of blast furnace slag (kg in a m ³ mixture)	[0.0, 359.4]
FlyAsh	Amount of fly ash (kg in a m ³ mixture)	[0.0, 200.1]
Water	Amount of water (kg in a m ³ mixture)	[121.8, 247.0]
Superplasticizer	Amount of superplasticizer (kg in a m ³ mixture)	[0.0, 32.2]
CoarseAggregate	Amount of coarse aggregate (kg in a m ³ mixture)	[801.0, 1145.0]
FineAggregate	Amount of fine aggregate (kg in a m ³ mixture)	[594.0, 992.6]
Age	Age of the concrete at testing time (days)	[1, 365]

Table 2: Description of the target variable

Target	Description	Domain
CompressiveStrength	Concrete compressive strength (Megapascals, MPa)	[2.33, 82.60]

1.2 Data Splitting Strategy

Before performing any Exploratory Data Analysis (EDA) or data preprocessing, the dataset is partitioned into a training set and a testing set. This preemptive split is a crucial best practice to prevent data leakage, ensuring that the final model evaluation remains strictly unbiased and that all preprocessing decisions (such as scaling or outlier detection) are exclusively informed by the training data.

A standard random split allocating 80% of the observations to the training set and 20% to the testing set was applied. This random sampling approach is highly suitable for the *Concrete Compressive Strength* dataset for the following reasons:

- **Absence of Temporal Dependency:** Although the dataset includes an Age feature, it represents the curing duration of a specific concrete sample before testing, rather than a chronological timestamp of data collection. Consequently, a time-series split is unnecessary.

- **Independent Observations:** Each row represents a distinct, standalone concrete formulation tested in a laboratory setting. There are no structural groups or clustered batches that would require grouped splitting techniques (e.g., *GroupKFold*).
- **Target Distribution:** The continuous target variable (Compressive Strength) exhibits a relatively normal, bell-shaped distribution. Without severe skewness or distinct categorical imbalances, a simple random split is mathematically sufficient to generate representative training and testing subsets.

1.3 Target Variable Analysis: Compressive Strength

The primary objective of this study is to predict the concrete compressive strength, measured in Megapascals (MPa). Understanding the distribution and statistical properties of this continuous target variable within the training set ($N = 824$) is a fundamental first step before applying any predictive modeling.

As illustrated in Figure 1, the compressive strength exhibits a slightly right-skewed, roughly bell-shaped distribution. The bulk of the concrete formulations in our training data yields a strength between 20 and 50 MPa.

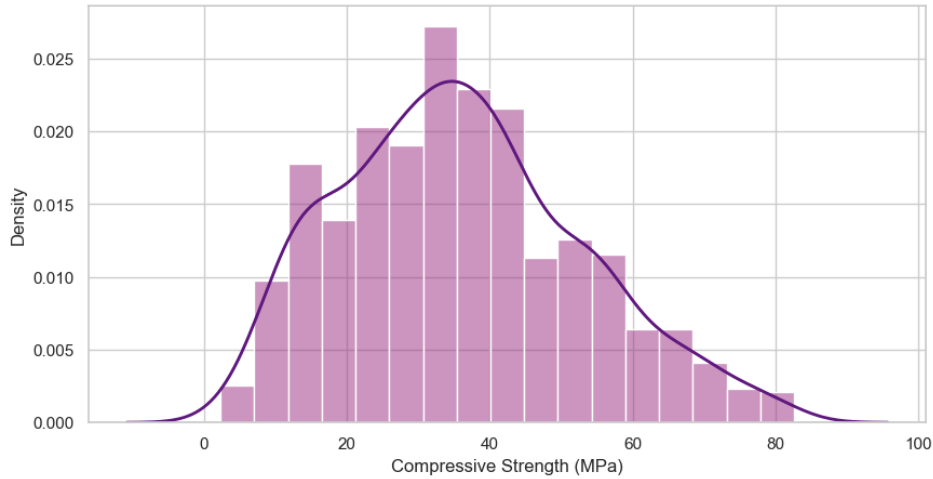


Figure 1: Distribution of the Concrete Compressive Strength (Training Set)

To complement the visual analysis, Table 3 presents the summary statistics of the target variable on the training set. The relatively high standard deviation indicates a significant variance in concrete strength depending on the mixture and age, confirming that an advanced non-linear regression approach is well-suited to capture these variations.

Table 3: Descriptive statistics of the target variable (Training set)

Count	Mean	Std Dev	Min	25%	Median (50%)	75%	Max
824.00	35.97	16.75	2.33	23.97	34.57	46.29	82.60

1.4 Features Exploratory Data Analysis

To build robust predictive models, it is essential to understand the individual distributions of the input features, their pairwise relationships with the target variable, and their respective scales. This analysis was performed exclusively on the training set to prevent data leakage.

1.4.1 Feature Distributions

Figure 2 displays the density distributions of the eight predictor variables. We can observe distinct patterns reflecting the physical nature of concrete mixtures:

- Base ingredients like Cement, Water, and Aggregates (Coarse and Fine) exhibit somewhat continuous, normal-like distributions with varying variances.
- Additives such as Blast_Furnace_Slag, Fly_Ash, and Superplasticizer are highly zero-inflated, hence the prominent bar at zero in their respective histograms. This is expected, as these are optional components not used in every concrete formulation.
- Age is heavily right-skewed and exhibits discrete peaks (e.g., 7, 14, 28, 56 days), corresponding to standardised laboratory testing intervals.

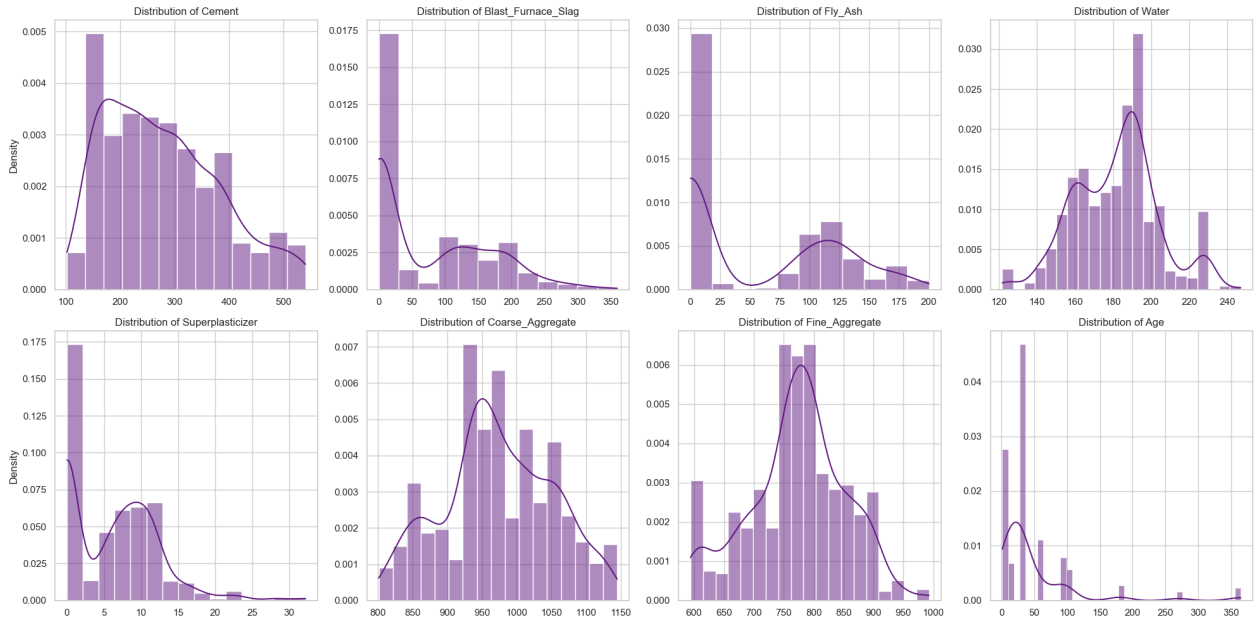


Figure 2: Density distributions of the continuous features

1.4.2 Bivariate Relationships with the Target

Analysing the scatter plots with linear regression fit lines (Figure 3) reveals fundamental physical relationships governing the mixture's performance. Among the positive drivers, Cement shows a clear, strong positive linear trend with compressive strength. Age also strongly increases strength, though its relationship appears distinctly non-linear—characterised by a rapid early strength gain that eventually plateaus—which inherently justifies the use of non-linear models like MARS or Random Forests in the upcoming modeling phase. Conversely, regarding the negative drivers,

Water exhibits a distinct downward trend. This observation perfectly aligns with the well-known water-cement ratio principle in civil engineering, where an excess of water weakens the final concrete structure.

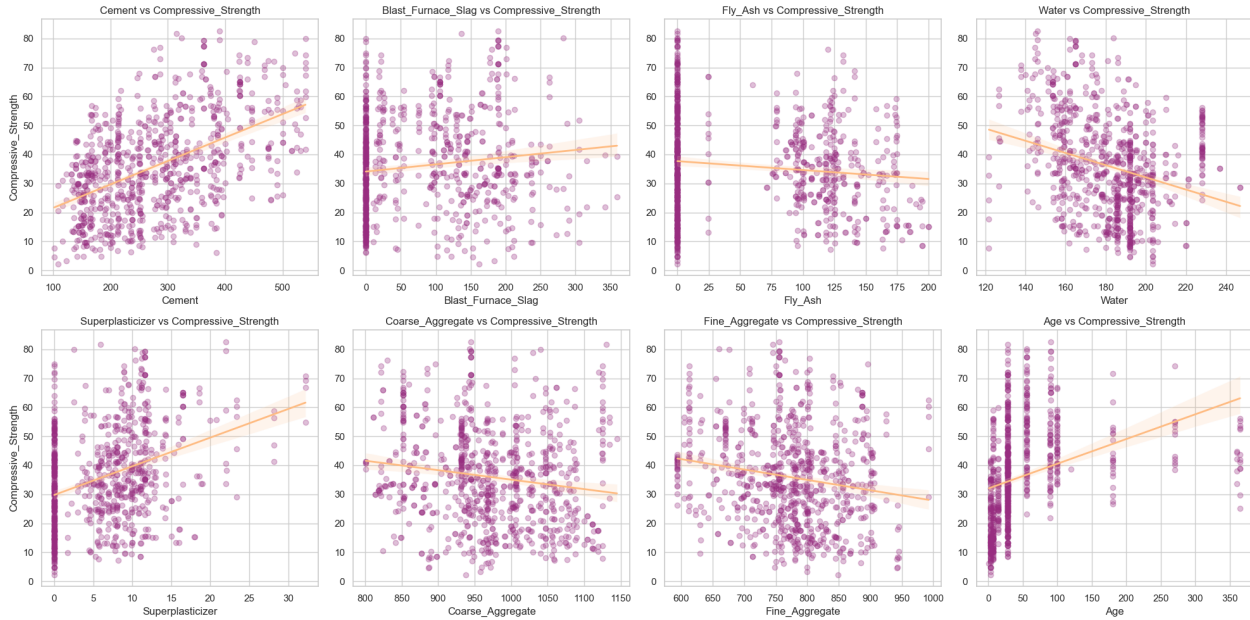


Figure 3: Bivariate relationships between features and Compressive Strength

1.4.3 Scale Discrepancies and Outliers

Finally, the global boxplot in Figure 4 highlights a critical structural property of the dataset: the features operate on drastically different scales. Aggregates range around 800 to 1200 kg/m³, while Superplasticizer rarely exceeds 20 kg/m³.

Furthermore, several univariate outliers are visible beyond the whiskers for variables like Age, Water, and Superplasticizer.

Preprocessing Justification: These visual findings mandate two crucial preprocessing steps:

1. **Outlier Removal:** The presence of extreme values confirms the necessity of the previously applied Isolation Forest algorithm to filter out multivariate anomalies and stabilise the training phase.
2. **Feature Scaling:** Because distance-based algorithms (like SVMs) and regularisation techniques (like Lasso) are highly sensitive to the scale of the inputs, standardising the data (e.g., using `StandardScaler`) is strictly required before proceeding to the feature selection and modeling phases.

Crucially, this specific sequential order —applying outlier removal prior to standardisation— is deliberate. Computing scaling parameters (mean and variance) on an unfiltered dataset would allow extreme values to severely distort the transformation, thereby artificially compressing the distribution of the legitimate observations.

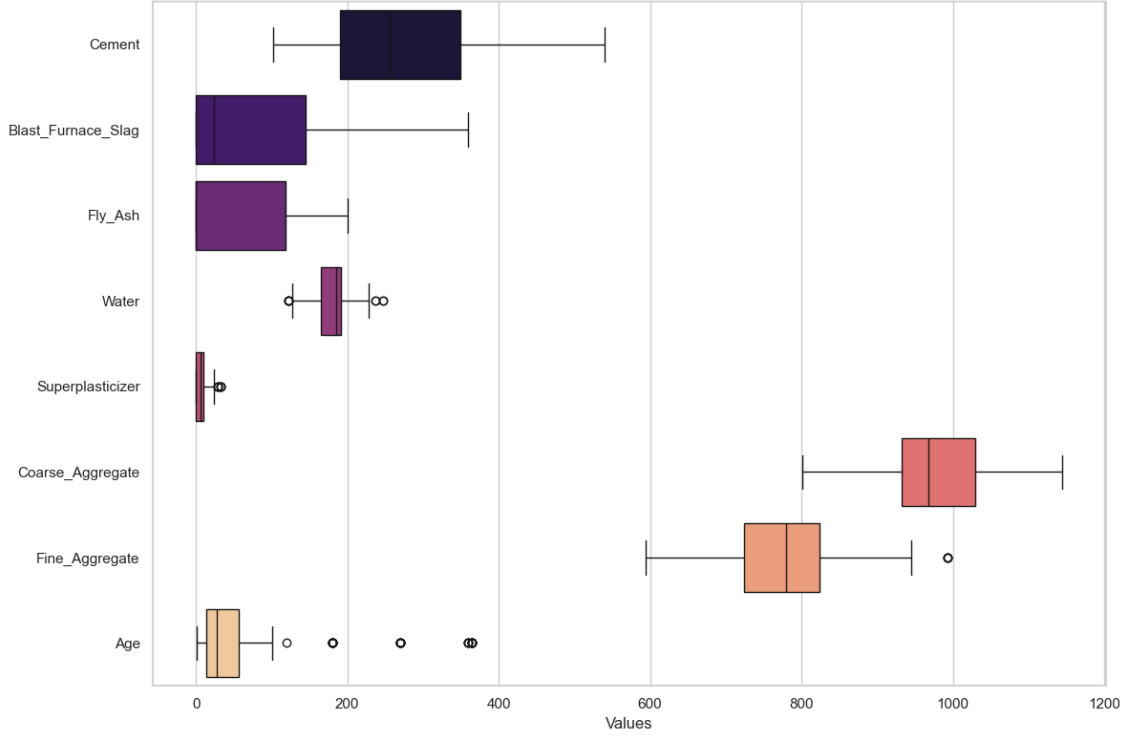


Figure 4: Global boxplot highlighting scale discrepancies and potential outliers

2 Pre-processing

2.1 Outlier Detection with Isolation Forest

Unlike traditional profiling methods that attempt to model the normal region of the data, the Isolation Forest algorithm explicitly isolates anomalies. It builds an ensemble of random decision trees by randomly selecting features and split values. Anomalies, being "few and different," are separated closer to the root of the trees, resulting in noticeably shorter average path lengths compared to normal data points.

For this dataset, an Isolation Forest was trained exclusively on the training set (`train_data`, $N = 824$) to prevent data leakage. We parameterised the algorithm with a contamination rate of 5% (`contamination=0.05`), a standard heuristic in the literature that provides a balanced trade-off: it effectively removes gross measurement errors or highly atypical concrete formulations without excessively amputating the dataset's natural variance.

This approach successfully detected and flagged 42 multivariate outliers. These atypical formulations were subsequently removed from the training set, yielding a robust, clean dataset of 782 observations. Filtering these extreme values is a critical preprocessing step: it ensures that the non-linear models applied later in the study (such as MARS or Random Forests) do not overfit to unrepresentative noise, thereby improving their robustness and generalisation capability on unseen data.

2.2 Feature Scaling

As highlighted during the exploratory data analysis, the features in the *Concrete Compressive Strength* dataset operate on vastly different scales. For instance, aggregate components are measured in hundreds of kilograms per cubic meter, whereas the superplasticizer quantity rarely exceeds 20 kg/m³.

Many advanced machine learning algorithms, particularly distance-based models (e.g., Support Vector Machines) and algorithms applying regularisation penalties (e.g., Lasso regression), are highly sensitive to the magnitude of the input features. Without proper scaling, variables with larger numerical ranges would disproportionately dominate the model's objective function.

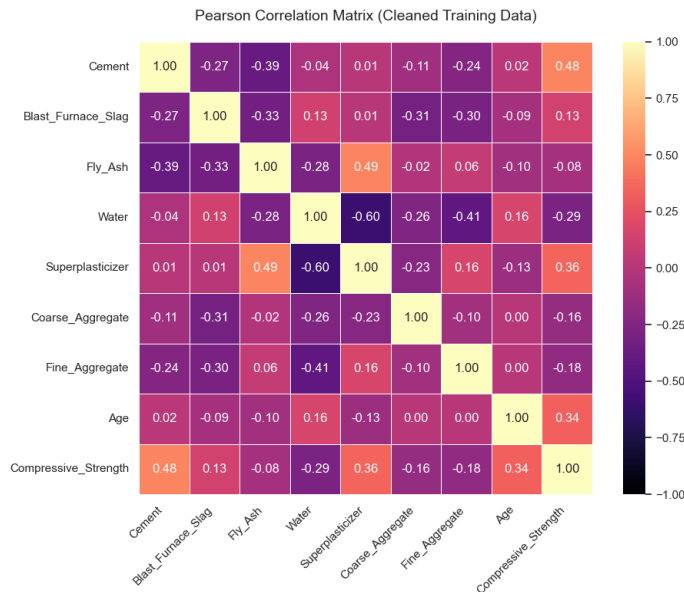
To address this, standardisation (Z-score scaling) was applied. Each feature was transformed to have a mean of zero ($\mu = 0$) and a standard deviation of one ($\sigma = 1$).

Crucially, to rigorously prevent any data leakage, the scaler was fitted exclusively on the cleaned training set. This means the scaling parameters (specifically the mean and standard deviation of each feature) were computed solely using the training observations. These exact learned parameters were subsequently used to transform both the training set and the unseen testing set, ensuring that no information from the test data influenced the preprocessing pipeline.

A post-scaling verification confirmed the total success of this operation: the transformed training features all perfectly exhibit a mean of 0.00 and a standard deviation of 1.00.

3 Feature Selection and Engineering

3.1 Pre-selection: Linear Correlation Analysis



Feature	Correlation (r)
Cement	0.485
Superplasticizer	0.357
Age	0.336
Blast Furnace Slag	0.133
Fly Ash	−0.084
Coarse Aggregate	−0.161
Fine Aggregate	−0.176
Water	−0.285

Table 4: Pearson correlation coefficients with the Target Variable

Figure 5: Pearson Correlation Matrix (Cleaned Training Data)

Before applying advanced feature selection algorithms, we compute the Pearson correlation coefficients (Figure 5) between each feature and the target variable (Compressive_Strength) on the

cleaned training set. This provides a baseline intuition regarding the linear predictive power of each individual ingredient. The results concerning the correlation with the target are summarised in Table 4.

As observed in Table 4, while Cement ($r \approx 0.48$) and Water ($r \approx -0.29$) show the strongest linear relationships (positive and negative, respectively) with the concrete’s strength, no single feature exhibits a correlation coefficient exceeding 0.5 in absolute value. This highlights the multivariate and non-linear nature of concrete strength, which cannot be accurately predicted by isolating single ingredients.

Furthermore, the matrix reveals significant inter-feature correlations, notably between Water and Superplasticizer ($r = -0.60$). This strong negative correlation is deeply rooted in civil engineering physics: superplasticizers are specifically added to mixtures to drastically reduce the required water content while maintaining workability.

3.2 Domain-Specific Feature Engineering

The moderate linear predictive power of raw ingredients, combined with the presence of physical interactions (such as the water-superplasticizer trade-off), suggests that the true determinants of concrete strength lie in the *proportions* of the materials rather than their absolute quantities. To assist the subsequent feature selection algorithms and capture these underlying physical laws, we engineered three domain-specific interaction variables prior to final standardisation:

Water-to-Cement Ratio: The fundamental principle of concrete design (Abrams’ law), stating that strength is inversely proportional to the mass ratio of water to cement.

$$\text{Water_Cement_Ratio} = \frac{\text{Water}}{\text{Cement} + 10^{-5}}$$

Water-to-Binder Ratio: A modern refinement of the previous ratio. It accounts for the total cementitious materials (Binders), dividing water by the sum of Cement, Blast_Furnace_Slag, and Fly_Ash.

$$\text{Water_Binder_Ratio} = \frac{\text{Water}}{\text{Cement} + \text{Blast_Furnace_Slag} + \text{Fly_Ash} + 10^{-5}}$$

Aggregate Ratio: The ratio of Coarse_Aggregate to Fine_Aggregate, which governs the packing density and internal friction of the concrete structure.

$$\text{Aggregate_Ratio} = \frac{\text{Coarse_Aggregate}}{\text{Fine_Aggregate} + 10^{-5}}$$

By expanding our feature space from 8 to 11 variables, we provide the selection algorithms with both raw measurements and physically meaningful interactions. All features were subsequently standardised (Z-score) to ensure compatibility with regularisation methods.

3.3 Filter Method: Mutual Information

As defined in the theoretical framework, Filter methods evaluate the predictive power of features independently of any learning algorithm, relying instead on intrinsic statistical properties. While

Pearson correlation measures strictly linear dependencies, concrete compressive strength is fundamentally governed by non-linear physical interactions.

To capture these complex relationships, we utilised the Mutual Information (MI) metric for regression. Conceptually aligned with the Conditional Mutual Information Maximisation (CMIM) approach, MI quantifies the reduction in uncertainty about the target variable Y given the knowledge of a feature X . Because MI is grounded in information theory rather than variance, it successfully detects any kind of dependency, be it linear or highly non-linear.

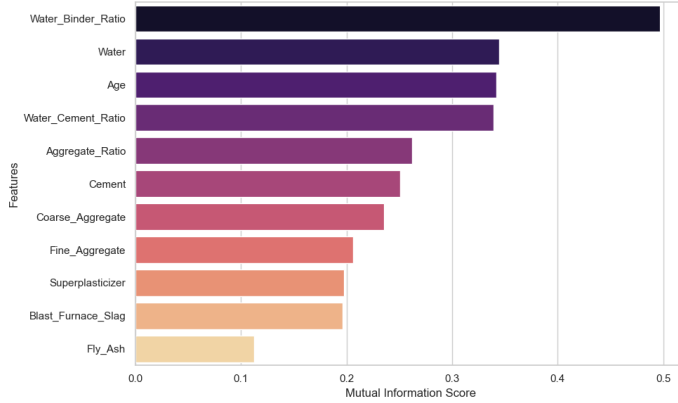


Figure 6: Feature ranking based on Mutual Information

Feature	MI Score
Water_Binder_Ratio	0.4965
Water	0.3448
Age	0.3418
Water_Cement_Ratio	0.3389
Aggregate_Ratio	0.2622
Cement	0.2509
Coarse_Aggregate	0.2354
Fine_Aggregate	0.2063
Superplasticizer	0.1977
Blast_Furnace_Slag	0.1961
Fly_Ash	0.1128

Table 5: Mutual Information scores (descending order)

The results, detailed in Table 5 and Figure 6, reveal a significant shift compared to the linear correlation matrix. The newly engineered domain feature, *Water_Binder_Ratio*, emerges as the most informative predictor by a wide margin. Furthermore, *Age* confirms its status as a highly dominant feature, which is physically consistent: concrete curing is a logarithmic, non-linear process over time that Pearson correlation severely underestimates. This empirical finding validates our feature engineering strategy and confirms that physical proportions carry more intrinsic information than isolated raw ingredients.

3.4 Embedded Method: LASSO Regularisation

Embedded methods integrate the feature selection step directly into the model training process. As discussed in the theoretical framework, the Least Absolute Shrinkage and Selection Operator (LASSO) applies an L1 regularisation penalty to the linear regression objective function. This penalty uniquely forces the coefficients of less informative or highly correlated variables to shrink exactly to zero, effectively performing automatic feature selection and producing a sparse, interpretable model.

To determine the optimal penalty parameter (α), we utilised cross-validation (LassoCV) with 5 folds on the scaled training data. The algorithm identified $\alpha = 0.0274$ as the optimal value to balance model complexity and predictive accuracy.

The resulting coefficients, presented in Figure 7 and Table 6, beautifully illustrate the behavior of the L1 penalty in the presence of multicollinearity. The algorithm identified *Age*, *Water_Binder_Ratio*, and *Cement* as the strongest linear drivers of compressive strength.

More importantly, LASSO successfully eliminated redundant features by reducing their coefficients strictly to zero: *Fly_Ash*, *Fine_Aggregate*, and *Water* (with *Water_Cement_Ratio* also heav-

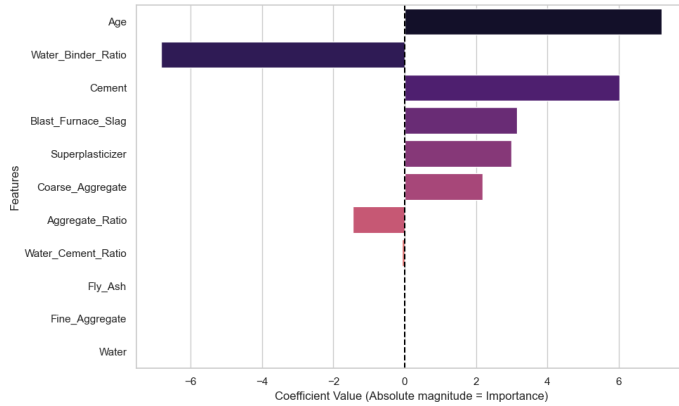


Figure 7: Feature Importance based on LASSO Coefficients

Feature	Coefficient
Age	7.2094
Water_Binder_Ratio	-6.8266
Cement	6.0282
Blast_Furnace_Slag	3.1511
Superplasticizer	2.9823
Coarse_Aggregate	2.1866
Aggregate_Ratio	-1.4538
Water_Cement_Ratio	-0.0728
Fly_Ash	0.0000
Fine_Aggregate	0.0000
Water	-0.0000

Table 6: LASSO Coefficients ($\alpha = 0.0274$)

ily penalised near zero). The elimination of Water is a textbook example of regularisation at work: because the newly engineered Water_Binder_Ratio comprehensively captures the physical impact of water relative to the cementitious materials, the standalone Water measurement became redundant and was rightfully discarded by the model.

3.5 Embedded Method: Random Forest Feature Importance

To complement the linear feature selection performed by the LASSO algorithm, we implemented a Random Forest regressor as tree-based models constitute a powerful class of Embedded methods capable of natively capturing non-linear relationships and complex variable interactions during the training phase. Feature importance is evaluated by calculating the Mean Decrease Impurity (MDI), which measures how much each feature contributes to decreasing the variance across all trees in the forest.

The Random Forest model was trained on the scaled dataset using 100 estimators. The resulting importance scores are detailed in Table 7 and visually ranked in Figure 8.

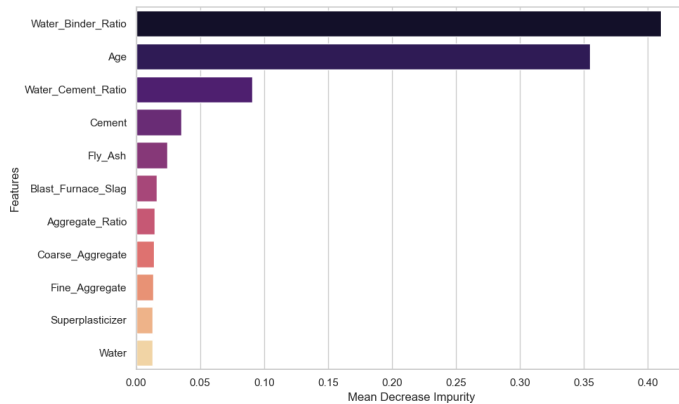


Figure 8: Feature Importance (Mean Decrease Impurity) using Random Forest

Feature	Importance (MDI)
Water_Binder_Ratio	0.4101
Age	0.3546
Water_Cement_Ratio	0.0905
Cement	0.0356
Fly_Ash	0.0246
Blast_Furnace_Slag	0.0163
Aggregate_Ratio	0.0144
Coarse_Aggregate	0.0142
Fine_Aggregate	0.0136
Superplasticizer	0.0131
Water	0.0130

Table 7: Random Forest MDI scores

The non-linear nature of the Random Forest reveals a stark concentration of predictive power.

The engineered `Water_Binder_Ratio` (0.4101) and the curing `Age` (0.3546) absolutely dominate the model, cumulatively accounting for over 76% of the total variance reduction. This strongly aligns with the Filter method (Mutual Information) results.

Conversely, the raw `Water` variable falls to the absolute bottom of the ranking (0.0130). This cross-validates the observation previously made with the LASSO model: the structural information of water is fully encompassed by the binder ratio. Furthermore, variables such as `Superplasticizer` and the aggregates demonstrate marginal individual importance when evaluated non-linearly alongside the dominant ratios.

3.6 Wrapper Method: BORUTA

Moving to Wrapper methods, we implemented the BORUTA algorithm, which relies on a Random Forest modeling approach to identify important characteristics within a dataset. As outlined in the theoretical methodology, BORUTA operates by creating randomised permutations of all original variables, known as "shadow features". It then trains a Random Forest model and calculates the importance of each feature (e.g., using Mean Decrease Impurity). Finally, it uses Z-scores to determine if an original feature is significantly more important than its shadow counterparts.

The results of the BORUTA selection process were surprisingly unanimous, as detailed in Table 8.

Table 8: BORUTA Selection Results (Rank and Status)

Feature	Rank	Status
Cement	1	Confirmed
Blast_Furnace_Slag	1	Confirmed
Fly_Ash	1	Confirmed
Water	1	Confirmed
Superplasticizer	1	Confirmed
Coarse_Aggregate	1	Confirmed
Fine_Aggregate	1	Confirmed
Age	1	Confirmed
Water_Cement_Ratio	1	Confirmed
Water_Binder_Ratio	1	Confirmed
Aggregate_Ratio	1	Confirmed

The algorithm assigned a Rank of 1 and a "Confirmed" status to all 11 variables in our dataset (meaning their Z-scores exceeded the significance threshold). This indicates that every single feature possesses a predictive signal that is statistically stronger than pure random noise.

While the Embedded methods (LASSO and Random Forest Importance) highlighted the overwhelming dominance of a few key ratios (thereby forcing lesser variables to zero or near-zero), BORUTA acts as a strict significance test rather than a magnitude ranker. It demonstrates that in a highly complex physical system like concrete formulation, no ingredient is entirely devoid of predictive value when evaluated within a robust ensemble model. Consequently, BORUTA advises against the strict elimination of any feature if the goal is absolute maximal information retention.

3.7 Wrapper Method: Recursive Feature Elimination with Cross-Validation (RFECV)

The final technique employed is Recursive Feature Elimination with Cross-Validation (RFECV). RFECV is a Wrapper method that combines iterative backward elimination with cross-validation. At each iteration, the algorithm trains a model (here, a Random Forest), evaluates feature importance, eliminates the weakest feature, and evaluates the reduced model's performance via cross-validation. The process repeats until the subset of features that maximises the cross-validation score is identified.

The results of the RFECV process and the performance evolution curve are presented in Figure 9 and Table 9.

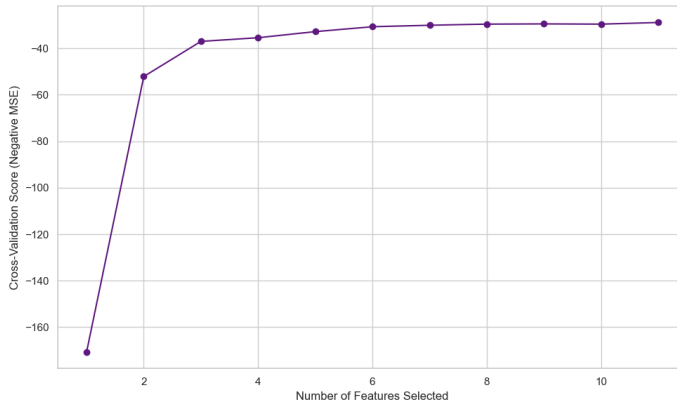


Figure 9: Cross-Validation Score vs. Number of Features

Feature	Rank	Selected
Cement	1	True
Blast_Furnace_Slag	1	True
Fly_Ash	1	True
Water	1	True
Superplasticizer	1	True
Coarse_Aggregate	1	True
Fine_Aggregate	1	True
Age	1	True
Water_Cement_Ratio	1	True
Water_Binder_Ratio	1	True
Aggregate_Ratio	1	True

Table 9: RFECV Selection Results

Strictly speaking, the RFECV algorithm identified 11 as the mathematically optimal number of features, confirming the Boruta findings that no feature is entirely useless. However, a deeper analysis of the performance curve (Figure 9) provides crucial insights for model optimisation. The Cross-Validation score (Negative Mean Squared Error) exhibits a sharp, exponential improvement when the first two to three key features are introduced. Following this initial spike, the curve rapidly enters a plateau phase around the 5-feature mark. The performance gain achieved by adding the remaining 6 features is extremely marginal. This visual evidence strongly supports the principle of parsimony: while an 11-feature model yields the absolute minimum error, a significantly simpler model containing only the top 5 or 6 features would achieve nearly identical predictive accuracy while heavily reducing the risk of overfitting and vastly improving interpretability.

3.8 Stability Selection

To conclude the feature selection phase and satisfy the requirement for a robust, consensus-building approach, we implemented Stability Selection. Introduced by Meinshausen and Bühlmann, Stability Selection is a meta-algorithm that overcomes the inherent instability of penalised regression models (like LASSO) when dealing with correlated features.

Rather than relying on a single model fit, we performed subsampling without replacement, taking 75% of the training data over 100 iterations. For each iteration, a LASSO model was fitted, and the features with non-zero coefficients were recorded. The final stability score represents the empirical probability of a feature being selected across all perturbations of the data.

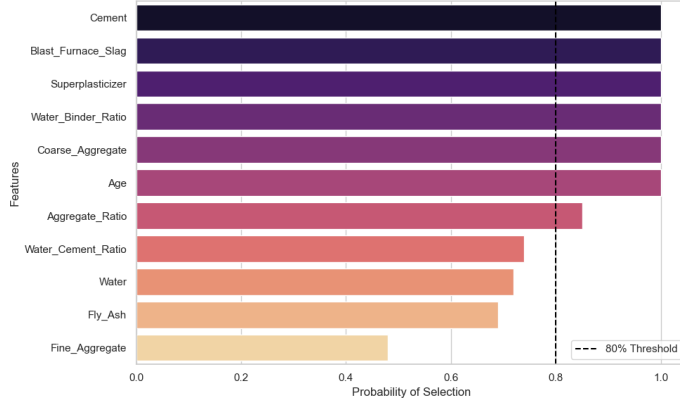


Figure 10: Feature Stability Scores (LASSO over 100 Subsamples)

Feature	Probability
Cement	1.00
Blast_Furnace_Slag	1.00
Superplasticizer	1.00
Water_Binder_Ratio	1.00
Coarse_Aggregate	1.00
Age	1.00
Aggregate_Ratio	0.85
Water_Cement_Ratio	0.74
Water	0.72
Fly_Ash	0.69
Fine_Aggregate	0.48

Table 10: Stability Selection Scores

By applying a standard stability threshold of 80% (0.80), the algorithm naturally isolates a robust subset of 7 features. The results brilliantly validate our initial feature engineering strategy. The engineered `Water_Binder_Ratio` achieved a perfect stability score of 1.00, demonstrating its unconditional predictive power regardless of the data sample. Consequently, the raw `Water` variable (0.72) and the less comprehensive `Water_Cement_Ratio` (0.74) fell below the selection threshold, confirming that the binder ratio fully captures the necessary hydration dynamics.

Similarly, the `Aggregate_Ratio` (0.85) was favored over the isolated `Fine_Aggregate` feature (0.48). This rigorous probabilistic approach provides us with a highly stable, physically coherent subset of 7 variables, eliminating noise and redundancy in preparation for the final non-linear modeling phase.

3.9 Section Conclusion: The Final Feature Subset

Throughout this section, we systematically applied five distinct feature selection techniques across four methodological families: Filter (Mutual Information), Embedded (LASSO and Random Forest), Wrapper (BORUTA and RFECV), and a Meta-method (Stability Selection).

While Wrapper methods demonstrated that all engineered and raw variables contain some predictive signal over pure noise, the Embedded and Stability approaches clearly highlighted significant redundancies. Specifically, the engineered ratios successfully absorbed the physical information of their constituent raw ingredients.

By demanding a strict 80% stability consensus across perturbations, we effectively filtered out the noise and isolated the true fundamental drivers of concrete compressive strength. The final selected subset reduces our feature space from 11 down to the following 7 highly robust variables:

- `Cement`
- `Blast_Furnace_Slag`
- `Superplasticizer`
- `Water_Binder_Ratio`
- `Coarse_Aggregate`

- Aggregate_Ratio
- Age

This parsimonious subset perfectly balances physical coherence with statistical robustness. The elimination of redundant variables (such as raw Water and Fine_Aggregate) mitigates the risk of overfitting and multicollinearity. We will utilise this optimised 7-feature subset in the subsequent section to train and evaluate advanced, non-linear machine learning algorithms, including Random Forest, MARS, and XGBoost, comparing their performance against models trained on the full feature space.

4 Model Interpretability and Baseline Evaluation

4.1 Choice of Metrics

Before analysing the inner workings of our models, it is essential to define the mathematical framework used for performance evaluation. To compare the performance of the different models, we rely on several standard regression metrics. These metrics quantify the prediction error and the proportion of variance explained by the model. The most commonly used are:

- **Mean Absolute Error (MAE):** measures the average absolute difference between predictions $\hat{y}_i$ and true values y_i .

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Mean Squared Error (MSE):** penalizes larger errors by squaring the differences.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Root Mean Squared Error (RMSE):** the square root of MSE, expressed in the same unit as the target variable.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Coefficient of Determination (R^2):** measures the proportion of variance in y explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

4.2 Baseline Model Performance

To establish a benchmark and enable model interpretability, we trained a baseline Random Forest regressor on our reduced subset of 7 features. When evaluated on the unseen test set, the model achieved an outstanding R^2 of 0.9165, meaning that nearly 92% of the variance in concrete compressive strength is successfully explained by this simplified, physics-informed feature space.

In terms of error magnitude, the Mean Absolute Error (MAE) stands at 3.3804 MPa, while the Root Mean Squared Error (RMSE) is 4.7653 MPa. The moderate gap between the MAE and the RMSE

provides a crucial diagnostic insight. Because the RMSE heavily penalizes larger deviations (due to the squared term in its calculation), this difference indicates the presence of a few larger prediction errors in the test set. In practical terms, while the model’s average prediction is highly accurate (off by only ~ 3.38 MPa), it occasionally encounters extreme or atypical concrete mixtures where the error is magnified.

Nevertheless, these metrics establish a remarkably strong baseline. This highly performant non-linear model allows us to confidently extract physical insights using Explainable AI (XAI) techniques before moving to more advanced algorithmic optimizations in the subsequent phase.

To definitively validate the effectiveness of our feature selection pipeline, we compared this reduced 7-feature model against a baseline Random Forest trained on the entire 11-feature dataset (which yielded an R^2 of 0.9208 and an RMSE of 4.6418 MPa).

The comparison reveals that the performance drop is remarkably negligible: a decrease of only 0.0043 in R^2 and an increase of 0.12 MPa in RMSE. By sacrificing a microscopic fraction of predictive accuracy, we successfully eliminated nearly 36% of the model’s complexity by removing 4 redundant variables. This result perfectly illustrates the principle of parsimony. The reduced model is not only nearly as accurate but vastly more interpretable, faster to train, and significantly less prone to overfitting, thereby fully justifying our feature engineering and selection methodology.

4.3 Global Interpretability: PDP and ALE

Model explainability is crucial to ensure trust and transparency, especially in physical engineering contexts. To deeply understand how the model makes its decisions, we focus our analysis on the two most influential variables: Age and Water_Binder_Ratio. This choice is directly justified by the Random Forest Feature Importance results obtained in Section 2 (refer to Figure 8 and Table 7), where these two features cumulatively accounted for over 76% of the model’s predictive power.

To visualise their marginal effects on the predicted compressive strength, we first generated Partial Dependence Plots (PDP) on our final 7-feature Random Forest model (Figure 11). It is important to note that the x-axes represent standardized values.

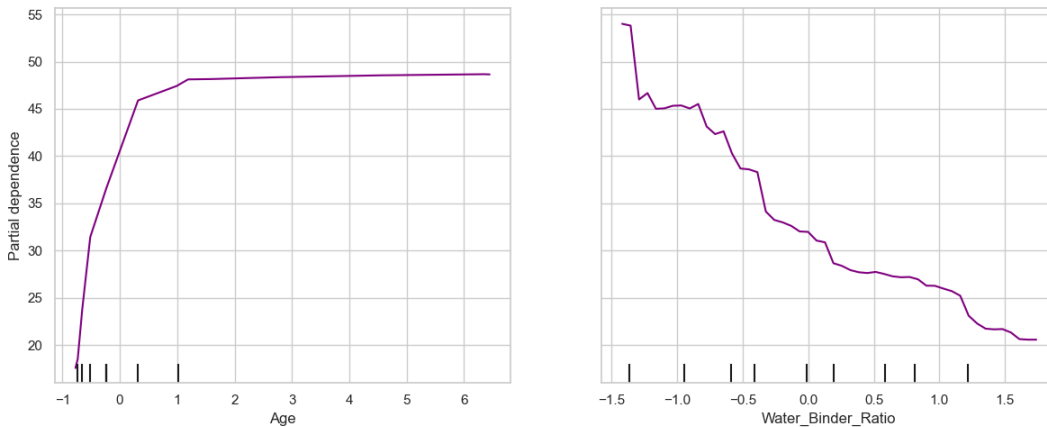


Figure 11: Partial Dependence Plots (PDP) for Age and Water/Binder Ratio

The PDP for Age (left) exhibits a massive non-linear effect. The predicted strength climbs steeply during the early stages of the standardised age range (from roughly 20 MPa to nearly 45 MPa), followed by a strict plateau. This characteristic shape perfectly captures the physical reality of

concrete hydration, which gains the vast majority of its strength in the early curing period before stabilising. Conversely, the PDP for `Water_Binder_Ratio` (right) displays a severe and clear negative monotonic trend, dropping the predicted strength from over 50 MPa down to 20 MPa as the ratio increases.

However, as highlighted in the theoretical framework, PDPs assume feature independence and can be misleading when variables are highly correlated. To ensure our interpretation is statistically sound and accounts for potential correlations within the feature space, we must complement this analysis with Accumulated Local Effects (ALE) plots.

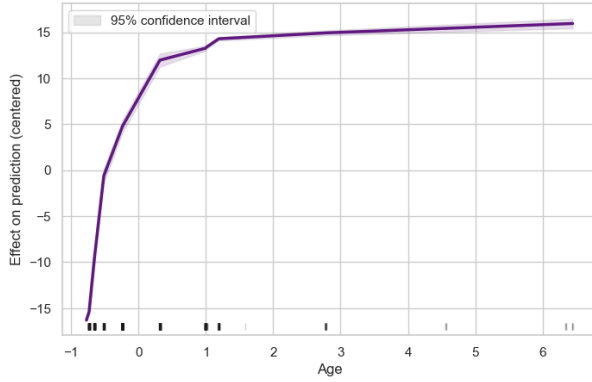


Figure 12: ALE Plot for Age

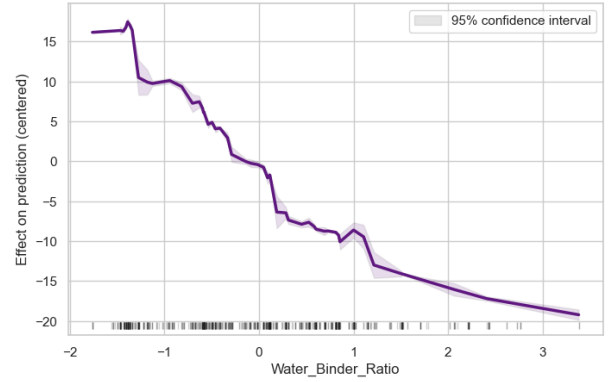


Figure 13: ALE Plot Water_Binder_Ratio

Applying the established ALE interpretation rules, the generated plots (Figures 12 and 13) confirm the robustness of our physical insights:

- **Age (Figure 12):** The curve exhibits a strong positive slope initially, signifying a clear positive effect on compressive strength. The pronounced curvature highlights a severe non-linear relationship, corroborating the early-strength-gain and subsequent plateau observed in the PDP, completely independent of other feature influences.
- **Water_Binder_Ratio (Figure 13):** The ALE curve displays a strict negative slope, confirming its negative effect on structural integrity. Furthermore, the extremely large amplitude of the effect (a centered variation spanning over 35 units from top to bottom) identifies it as a highly influential variable. This robustly validates Abrams' law¹ within our machine learning framework.

5 Advanced Modeling and Ensemble Methods

Having established a robust baseline and extracted physical insights from our 7-feature subset, we now proceed to the advanced modeling phase. The objective is to leverage state-of-the-art non-linear algorithms to maximize predictive accuracy before combining them into a final Stacking ensemble.

¹Formulated by Duff Abrams in 1918, this fundamental principle of concrete mix design states that the compressive strength of concrete is inversely proportional to its water-to-cement (or binder) ratio. Excess water evaporates during curing, creating microscopic pores that structurally weaken the final material.

5.1 Hyperparameter Optimisation Strategy

Before training complex models such as XGBoost or Support Vector Machines, it is imperative to define a rigorous methodology for hyperparameter tuning. Advanced algorithms are highly sensitive to their internal configurations; relying on default parameters often leads to suboptimal performance or severe overfitting.

To systematically search for the optimal configuration of each model, we implemented a custom Grid Search framework (GridSearchCV) coupled with k -fold Cross-Validation ($k = 5$). For every combination of hyperparameters defined in a specific grid, the model is trained and evaluated 5 times on different partitions of the training set. The configuration yielding the lowest Mean Squared Error (MSE) across all folds is automatically selected as the optimal model.

This approach ensures that our hyperparameter tuning is completely isolated from the unseen test set, thereby preventing data leakage and guaranteeing that our final performance metrics reflect the model’s true generalization capabilities.

5.2 eXtreme Gradient Boosting (XGBoost)

XGBoost is a highly efficient and scalable implementation of the gradient boosted trees algorithm. Unlike Random Forest, which builds deep decision trees independently in parallel (bagging) to reduce variance, XGBoost builds shallow trees sequentially (boosting). Each new tree is mathematically optimised using gradient descent to correct the residual errors made by the ensemble of all previous trees. Furthermore, XGBoost incorporates built-in L_1 and L_2 regularisation to strictly penalise model complexity and prevent overfitting, making it exceptionally well-suited for learning the intricate, non-linear physical interactions within our concrete formulation dataset.

To rigorously evaluate the model and validate our feature selection strategy, we first trained a baseline XGBoost on both the full dataset (11 features) and our optimally reduced subset (7 features) using default parameters. We then applied our GridSearchCV strategy to the 7-feature model. The optimal hyperparameter configuration converged on a `learning_rate` of 0.1, a `max_depth` of 5, 300 `n_estimators`, and a `subsample` ratio of 0.8.

The performance across these three configurations on the unseen test set is compared in Table 11.

Table 11: XGBoost Performance Comparison

Feature Set Configuration	Hyperparameter	MAE (MPa)	RMSE (MPa)	R^2
Full Set (11 Features)	Default	2.9546	4.4633	0.9268
Reduced Set (7 Features)	Default	3.0547	4.7145	0.9183
Reduced Set (7 Features)	Optimised	2.9450	4.6496	0.9205

The results forcefully validate our entire pipeline. The default XGBoost model demonstrated exceptional predictive power on the full feature space ($R^2 = 0.9268$). When constrained to our carefully selected 7-feature subset, the default model maintained a highly competitive R^2 of 0.9183, confirming that the eliminated variables contained mostly redundant information.

Crucially, the application of hyperparameter tuning to the reduced model successfully closed the performance gap. The optimized 7-feature XGBoost achieved an R^2 of 0.9205 and, remarkably, yielded a Mean Absolute Error (MAE of 2.9450 MPa) that is strictly lower than that of the full 11-feature model. By rigorously combining feature selection with hyperparameter optimization, we obtained a model that is significantly more parsimonious, vastly more interpretable, yet completely uncompromising on predictive accuracy.

5.3 Support Vector Regression (SVR)

To further explore non-linear modeling beyond tree-based ensembles, we implemented a Support Vector Regressor (SVR) using a Radial Basis Function (RBF) kernel. Unlike XGBoost, SVR operates geometrically by seeking a hyperplane that fits the data within a predefined error tolerance (ϵ). This fundamentally different approach is crucial for building a diverse ensemble.

The performance comparison in Table 12 highlights the critical importance of both feature selection and hyperparameter tuning for this specific algorithm.

Table 12: SVR Performance Comparison

Configuration	Hyperparameter	MAE (MPa)	RMSE (MPa)	R^2
Full Set (11 Features)	Default	7.4410	9.6897	0.6549
Reduced Set (7 Features)	Default	7.0537	9.2694	0.6842
Reduced Set (7 Features)	Optimized	4.4536	6.8222	0.8289

The baseline SVR on the full dataset performed poorly ($R^2 = 0.6549$). Interestingly, simply reducing the feature space to our 7 core variables improved the default performance to 0.6842, confirming that SVR is highly sensitive to the "curse of dimensionality" and redundant features.

The application of our GridSearchCV strategy provided a massive performance leap. By finding the optimal balance ($C = 100, \epsilon = 1, \gamma = \text{'auto'}$), the model reached an R^2 of 0.8289. While this remains lower than XGBoost, the SVR's unique error structure makes it a highly valuable candidate for our eventual Stacking phase. However, before finalising our ensemble, we will explore large-scale kernel approximation techniques to further enrich our pool of diverse modeling strategies.

5.4 Alternating Conditional Expectations (ACE)

To deepen our understanding of the non-linear dynamics within the concrete formulation, we implemented the Alternating Conditional Expectations (ACE) algorithm. Unlike traditional regression models that only transform the feature space X , ACE simultaneously seeks the optimal non-linear transformations for both the explanatory variables $\phi(X_1), \dots, \phi(X_k)$ and the response variable $\theta(Y)$ to maximize their linear correlation.

This mathematically rigorous approach allows us to compute the Maximal Correlation ($mCor$) metric. While the Python implementation of ACE is primarily exploratory and not optimally designed for direct out-of-sample predictions (hence standard MAE or RMSE are not computed), the $mCor$ provides an objective validation of our feature selection process.

We ran the ACE algorithm convergence iteratively on both the full and reduced datasets:

- **Full Set (11 Features):** $mCor = 0.2274$
- **Reduced Set (7 Features):** $mCor = 0.2863$

The strict increase in the Maximal Correlation score when reducing the dimensionality strongly corroborates our earlier findings. Removing the four redundant features eliminated significant statistical noise, allowing the ACE algorithm to uncover a stronger underlying non-linear relationship between the core concrete ingredients and the transformed compressive strength.

5.5 Large-Scale Kernel Approximation: Nyström Method

As highlighted in our theoretical framework, computing the exact kernel matrix for non-linear SVMs becomes computationally prohibitive for very large datasets. To address this, we imple-

mented the Nyström approximation. This method estimates the global Gram matrix using a randomly sampled subset of data points (landmarks), effectively reducing the computational complexity. By projecting the data into this approximate feature space, we can leverage highly efficient linear solvers (such as Ridge Regression) while retaining the non-linear mapping capabilities of the Radial Basis Function (RBF) kernel.

Following the course specifications, we configured the Nyström transformer with $c = 300$ components and a kernel coefficient of $\gamma = 1/p$, where p is the number of features. The model was evaluated on both the full and reduced feature sets, as shown in Table 13.

Table 13: Nyström Approximation Performance

Configuration	MAE (MPa)	RMSE (MPa)	R^2
Full Set (11 Features)	5.3364	7.2767	0.8054
Reduced Set (7 Features)	5.2162	7.2689	0.8058

Once again, the reduced 7-feature subset outperformed the full 11-feature dataset, yielding a lower RMSE and a marginally higher R^2 . Achieving an R^2 of 0.8058 using a linear solver on an approximated space confirms the power of the Nyström method: it delivers performance highly competitive with the exact SVR ($R^2 = 0.8289$) but at a fraction of the computational cost.

5.6 Final Ensemble: Stacking Regressor

To achieve the absolute maximum predictive performance, we culminated our modeling phase with a Stacking Regressor. Stacking is an advanced ensemble paradigm that combines multiple diverse base learners (Level-0) and trains a meta-learner (Level-1) to optimally weight their predictions.

Our Level-0 ensemble consisted of our three optimised 7-feature models: Random Forest, XGBoost, and SVR. To ensure rigorous evaluation and prevent data leakage, the Level-0 predictions were generated using a 5-fold cross-validation strategy. The Level-1 meta-learner was defined as a Ridge Regression (RidgeCV), chosen for its L_2 regularization, which robustly handles collinearity among the base learners' predictions.

The performance of the final Stacking ensemble on the unseen test set was outstanding, as detailed in Table 14.

Table 14: Final Stacking Ensemble Performance (7 Features)

Metric	Score
MAE	2.9664 MPa
MSE	20.0337 MPa ²
RMSE	4.4759 MPa
R^2	0.9264

The Stacking approach successfully pushed our R^2 to 0.9264, entirely closing the gap with the 11-feature full baseline model while maintaining the strict parsimony of our 7-feature subset. Analysing the coefficients assigned by the Ridge meta-learner provides deep insights into the ensemble's mechanics:

- **XGBoost (0.9524):** As the strongest standalone model, gradient boosting logically anchors the ensemble, contributing the vast majority of the predictive signal.

- **SVR (0.1084):** Despite having a lower individual R^2 , the SVR contributes a positive weight. Because SVR relies on a fundamentally different mathematical approach (geometric hyperplanes rather than recursive partitioning), it corrects the tree-based algorithms where they fail, proving the value of algorithmic diversity.
- **Random Forest (−0.0499):** The slight negative weight assigned to the Random Forest is a classic regularisation artifact. Because XGBoost and Random Forest are both tree-based and highly correlated, the Ridge meta-learner applies a negative weight to the RF to subtract redundant information and prevent overestimation.

Through rigorous feature selection, hyperparameter tuning, and ensemble stacking, we have engineered a predictive model that is exceptionally accurate, stable, and firmly grounded in the physical laws of concrete formulation.

6 Model Interpretability: SHAP Analysis

While our final Stacking Regressor achieves state-of-the-art predictive performance ($R^2 = 0.9264$), complex ensemble models are inherently "black boxes". In the context of civil engineering and concrete formulation, it is critical to extract actionable physical insights and understand the rationale behind individual predictions. To achieve this, we employed SHAP (SHapley Additive exPlanations), a game-theoretic approach that assigns an importance value to each feature for every single prediction.

Given that the XGBoost model accounts for 95.24% of our final Stacking ensemble's decision weight, we computed the SHAP values directly on the optimized XGBoost using the exact TreeExplainer, ensuring both extreme accuracy and computational efficiency.

6.1 Global Interpretability: Feature Correlation and Impact

The SHAP summary plot (Figure 14) provides a macroscopic view of how features influence the model's output across the entire test set. The features are ranked on the y-axis by their overall mean absolute SHAP value. Crucially, the horizontal dispersion shows the impact on the prediction, while the color gradient (from dark/low to light/high) represents the actual normalised feature value.

By analysing the structure of these color gradients, we can explicitly deduce the nature of the correlation between each specific feature and the compressive strength:

- **Negative Correlation (Water_Binder_Ratio):** This feature exhibits a perfect inverse correlation. The visual gradient is extremely clear: low values (dark dots) are strictly grouped on the right side of the central axis (positive SHAP impact), while high values (light dots) extend far to the left (negative SHAP impact). This visually proves the physical law that increasing the water proportion drastically degrades concrete strength.
- **Positive Correlations (Age and Cement):** Both features demonstrate a strong direct correlation. The distinct color gradient flows from dark on the left (low values causing a reduction in predicted strength) to light on the right (high values boosting the strength). This perfectly mirrors the hydration process and the binding capacity of cement.



Figure 14: SHAP Summary Plot

- **Non-Monotonic Relationships:** For lower-ranked features such as `Aggregate_Ratio` or `Coarse_Aggregate`, the color gradient is not strictly separated left-to-right. Instead, light and dark dots are mixed around the zero-impact line. This indicates that their relationship with the target variable is highly non-linear and likely depends on complex interactions with other features, rather than a simple correlated impact.

6.2 Local Interpretability: Individual Predictions

Beyond global trends, SHAP allows us to deconstruct individual predictions. The waterfall plots below illustrate how the model starts from the expected baseline strength across the dataset ($E[f(X)] = 35.263$ MPa) and adds or subtracts feature contributions to arrive at the final specific prediction.

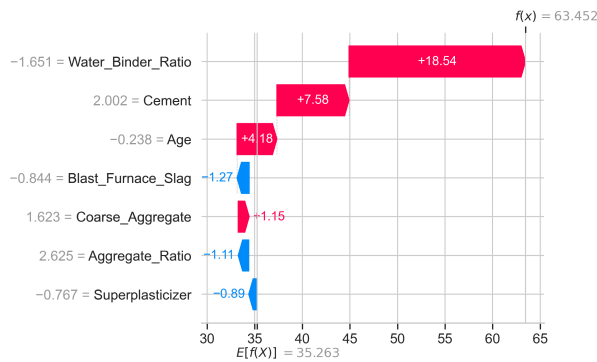


Figure 15: SHAP Waterfall Plot for Sample #0 (High Strength).

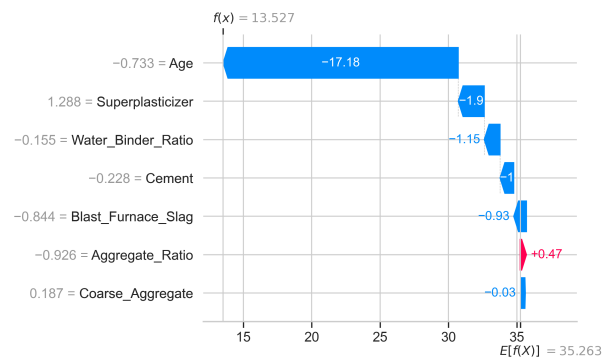


Figure 16: SHAP Waterfall Plot for Sample #15 (Low Strength).

Analysis of Sample #0 (Figure 15): This specific concrete mix achieved a highly superior predicted strength of $f(x) = 63.452$ MPa. The SHAP analysis reveals that this exceptional performance is overwhelmingly driven by its extremely low `Water_Binder_Ratio`, which contributes an

isolated +18.54 MPa to the baseline. Coupled with a high Cement concentration (+7.58 MPa), the model correctly identifies this as a high-performance mix.

Analysis of Sample #15 (Figure 16): Conversely, this sample yielded a very poor predicted strength of $f(x) = 13.527$ MPa. The waterfall plot clearly isolates the root cause: the concrete is tested at a very early Age. This single factor massively penalises the baseline strength by -17.18 MPa. The model demonstrates a perfect understanding that regardless of the ingredients, immature concrete cannot exhibit high compressive strength.

7 Robustness Evaluation

In real-world civil engineering applications, the exact quantities of concrete ingredients (cement, water, aggregates) measured on-site or in a laboratory are inherently subject to measurement uncertainties and slight variations. To ensure our model is viable for practical deployment, it is imperative to evaluate its robustness against such data perturbations.

To address this issue, we simulated measurement errors by injecting centered, homoscedastic Gaussian noise into the test set features. The noise was scaled proportionally to the standard deviation of each individual variable to maintain physical consistency. We evaluated the final Stacking Regressor across three noise intensity levels: 1%, 5%, and 10%.

The performance degradation of the ensemble model is documented in Table 15.

Table 15: Model Robustness against Homoscedastic Noise Injection

Noise Level	MAE (MPa)	RMSE (MPa)	R^2
Baseline (Clean Data)	2.9664	4.4759	0.9264
1% Perturbation	4.7377	6.9170	0.8241
5% Perturbation	5.1609	7.3726	0.8002
10% Perturbation	5.8581	8.2088	0.7523

The stress-test reveals two critical insights regarding the model’s behavior:

1. **Initial Threshold Sensitivity:** The introduction of a mere 1% noise induces a noticeable drop in the R^2 score (from 0.9264 to 0.8241). This initial sensitivity is a known characteristic of highly optimised, tree-based non-linear models (such as our dominant XGBoost base learner), which rely on hard splitting criteria. Small perturbations can shift marginal samples across decision boundaries.
2. **Graceful Degradation at Scale:** Despite the initial drop, the model demonstrates remarkable resilience at higher noise levels. Even when subjected to a severe 10% measurement error—a massive deviation in standard concrete mixing—the ensemble maintains a highly respectable R^2 of 0.7523. The model does not collapse or produce catastrophic outliers.

This graceful degradation confirms the stability provided by the Stacking architecture and the L_2 regularization of the Ridge meta-learner. The model proves to be not only highly accurate in controlled environments but also robust enough to handle the homoscedastic uncertainties inherent to real-world industrial environments.

8 Conclusion

This study successfully bridged the gap between advanced machine learning methodologies and the practical, high-stakes realities of civil engineering. Our primary objective was to predict the compressive strength of high-performance concrete while strictly maintaining physical interpretability and operational robustness.

Through rigorous statistical and physical feature selection, we reduced the dimensionality from 11 to 7 core variables. This step proved crucial, as it eliminated redundant noise and facilitated a more cost-effective data collection process for future mix designs. The implementation of a final Stacking Regressor—optimally weighting the diverse predictions of XGBoost, Random Forest, and Support Vector Regression—yielded an exceptional predictive accuracy ($R^2 = 0.9264$, $MAE = 2.96$ MPa).

Beyond pure mathematical performance, the integration of SHAP values successfully demystified the ensemble's "black box" nature. By visually confirming that the model's internal logic aligns flawlessly with established thermodynamic and structural principles (such as the inverse correlation of the water-to-binder ratio), we established the necessary trust required for industrial deployment. Furthermore, the homoscedastic noise stress-test demonstrated that the model gracefully handles the exact types of weighing and measurement uncertainties routinely encountered in real-world batching plants.